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碩士論文

An Orthogonal Procrustes Approach to Perturbation of Singular Value Decomposition (SVD) Problem and a Derivation of the Low-Rank Modification of SVD from Existing Literature

奇異值分解擾動問題：一種正交 Procrustes 方法暨奇異值分解的低秩修正方法之文獻整理與推導

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Abstract

This study consists of two topics: the low-rank modification of singular value decomposition (SVD) and the perturbation analysis of SVD. For the former, we organize a complete algorithm for low-rank SVD modification problem based on the works of Brand [2], Gandhi et al. [5], Golub [6], and Wilkinson [20].

The main contribution of this study lies in another topic, namely, the perturbation analysis of SVD. To the best of our knowledge, we propose an original approach that draws on the orthogonal Procrustes problem and incorporates Taylor-like perturbation expansions. This leads to an SVD approximation for $\tilde{A}\big|_{N(A)^\perp}$ (perturbed matrix $\tilde{A}$ restricted to the orthogonal complement of the null space of the original matrix A) that preserves the SVD-structure and costs only approximately $nmr + O\left((n \vee m)r^2\right)$ operations, where $A \in \mathbb{R}^{n\times m}$ and $\mathrm{rank}(A) = r$. Numerical experiments suggest that the residual term induced from the SVD approximation for $\tilde{A}\big|_{N(A)^\perp}$ provided in this study exhibits first-order convergence, i.e., of order $O(\|\Delta A\|)$.

Keywords: singular value decomposition, low-rank SVD modification, SVD perturbation analysis, orthogonal Procrustes problem, Taylor-like perturbation expansion, *sin* Θ theory

摘要

本研究主要涵蓋兩個主題：其一爲奇異值分解 (SVD) 的低秩更新問題，其二爲 SVD 的擾動分析。針對前者，本文依據 Brand [2]、Gandhi et al. [5]、Golub [6] 及 Wilkinson [20] 等文獻，整理出一套完整的低秩更新演算法。

除了 SVD 的低秩更新問題，本論文更加著墨於奇異值分解擾動問題。據我們所知，本研究提出一種具原創性的處理方法，其結合了用於處理正交 Procrustes 問題的技術與 Taylor-like 擾動展開式的概念，從而推導出擾動矩陣 $\tilde{A}$ 在原矩陣 A 之零空間的正交補集的奇異值分解估計式。該估計式除了完整地保持奇異值分解的結構外，只需花費大約 $nmr + O\left((n \vee m)r^2\right)$ 的計算量 (若 $A \in \mathbb{R}^{n\times m}$ 且 A 的秩是 r)。該估計式之誤差的收斂速度經數值實驗驗證爲 $O(\|\Delta A\|)$，屬於一階收斂。

關鍵字：奇異值分解、奇異值分解的低秩更新、奇異值分解的擾動分析、正交 Procrustes 問題、Taylor-like 擾動展開式、$sin\,\Theta$ 理論

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Notations

Symbol	Description
$\mathbb{N}$	The set consists of positive integers.
$A \in \mathbb{R}^{n \times m}$	A is a real matrix with n rows and m columns.
$A_{i,j}$	The entry of matrix A at the i-th row and j-th column.
A^T	The transpose of the matrix A.
$\det(A)$	The determinant of the matrix A.
diag	Diagonal matrix. For example, $\mathrm{diag}(1,2) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$.
$\mathrm{Tr}(A)$	The trace of the matrix A.
$N(A)$	The null space of A.
I	The identity matrix.
$\perp$	Orthogonal complement.
$\mathrm{rank}(A)$	The rank of the matrix A.
$A := B$	A is defined as B.
$A \triangleq B$	A is denoted as B.
$A \overset{i}{=} B$	A equals B up to terms of order i; higher-order terms are omitted.
$n \wedge m$	The minimum of n and m.
$n \vee m$	The maximum of n and m.
$F\big\vert_S$	The function F restricted to the set S. I.e., $F\big\vert_S : x \in S \mapsto F(x)$.
$\lVert \cdot \rVert_F$	The Frobenius norm.
$\langle A, B \rangle_F$	The Frobenius inner product $\sum_{i,j} A_{i,j} B_{i,j}$.

Chapter 1

Introduction

1.1 Problem Description

This thesis is concerned with two problems. The first problem concerns the low-rank modification of SVD (refer to [2]). The second problem, rigorously discussed in Chapter 3, concerns the perturbation of SVD (see [7, 8, 9, 10, 12, 15, 17, 18, 19, 21]). Although these two problems are related to SVD, they are fundamentally different, both in nature and in the techniques used to address them. In the rest of this section, we provide brief formulations of the two problems to indicate the focus of our study, while detailed developments are presented in Chapters 2 and 3.

Let $A \in \mathbb{R}^{n\times m}$. Suppose that $A = U\Sigma V^T$ is an SVD of the matrix A. The low-rank modification problem aims to derive, rather than directly evaluate, an SVD of the matrix $\hat{A} := A + L$, where L is a low-rank matrix; for example, $\operatorname{rank}(L) \leq O(\sqrt{n \wedge m})$ (see [2]). On the other hand, the perturbation of SVD problem aims to obtain, rather than directly evaluate, an SVD of $\tilde{A} := A + \Delta A$, where the perturbation ΔA is assumed to be small in norm ($\|\Delta A\| \ll \|A\|$ in a rough sense). However, due to theoretical difficulties and numerical limitations, one often resorts to deriving approximations of the singular values, singular vectors, or invariant subspaces of modified matrix $\hat{A}$ or perturbed matrix $\tilde{A}$ (see [6, 7, 8, 9, 10, 12, 13, 17, 18, 19, 21]).

1.2 Literature Review

This study covers two topics: low-rank modification of the SVD and perturbation of the SVD. The primary focus in this thesis lies on the latter. As a result, the literature review on low-rank modification is relatively limited. The rest of this section is devoted first to a review of the literature on low-rank modification, and then to that on perturbation of the SVD.

The low-rank modification problem aims to derive an SVD of the modified matrix $\hat{A} = A + L$. And after some derivations (refer to Brand [2]), it suffices to compute an SVD of the matrix K, where K is a low-rank modification of a diagonal matrix. Nevertheless, when the matrix A is not low-rank, the matrix K is not small in size, then computing its SVD directly entails a high computational cost. Since the matrix K is a low-rank modification of a diagonal matrix and the theory of rank-one update of the symmetric form $\Sigma + uu^T$, where Σ is a diagonal matrix and u is a vector, is well-developed (refer to Baglama [1]; Gandhi et al. [5]; Golub [6]; Wilkinson [20]), we adopt the strategy of successively applying rank-one updates to compute the SVD of the matrix K. Note that in the algorithm organized in this study, we avoid directly considering the SVD of the nonsymmetric form $\Lambda + uv^T$ (cf. Baglama [1]). For each rank-one update, we first transform the nonsymmetric type $\Lambda + uv^T$ into the symmetric type $\hat{\Lambda} + \hat{u}\hat{u}^T$ (see Gandhi et al. [5]), and then address the symmetric form $\hat{\Lambda} + \hat{u}\hat{u}^T$ (see Golub [6]; Wilkinson [20]). The details of the organized algorithm for the low-rank modification problem is presented in the Chapter 2.

As for the perturbation of SVD problem, to the best of our knowledge, existing methods can be broadly classified into two categories: one is the so-called *sin* Θ theory (see Dopico [4]), which provides norm or angle bounds for invariant subspaces rather than for individual singular vectors (cf. Jia and Stewart [8]; Stewart [15]); and the other approximates singular values and vectors through Taylor-like expansions with respect to the perturbation. Such Taylor-like expansions can be further classified into two types: first-order approximations (cf. Li and Vaccaro [9]; Liu et al. [10]) and second-order approximations (cf. Gratton and Tshimanga [7]; Vaccaro [17]; Vu et al. [18]; Xu et al. [21]).

Let ΔA be a small perturbation of the matrix A, such that $\|\Delta A\| \ll \|A\|$, and define the perturbed matrix $\tilde{A} := A + \Delta A$. Let σ_i and v_i denote the i-th largest singular value and the corresponding right singular vector of A, and let $\tilde{\sigma}_i$ and $\tilde{v}_i$ denote the corresponding perturbed singular value and singular vector of $\tilde{A}$, respectively. Then there exist a real number $\Delta\sigma_i$ and

a vector Δv_i such that $\tilde{\sigma}_i = \sigma_i + \Delta\sigma_i$ and $\tilde{v}_i = v_i + \Delta v_i$. Intuitively, one may expect both $|\Delta\sigma_i|$ and $\|\Delta v_i\|$ to be small, given that $\|\Delta A\| \ll \|A\|$. However, this is not always the case for singular vectors.

According to Weyl [19], we have $|\Delta\sigma_i| \leq \|\Delta A\|_2$ (see also Mirsky [12]). Thus, the perturbation $\Delta\sigma_i$ is guaranteed to be small in magnitude whenever $\|\Delta A\|$ is sufficiently small. It is reasonable to approximate the perturbed singular value $\tilde{\sigma}_i$ using Taylor-like perturbation expansions. For example, it is straightforward to show that the following expression provides a first-order approximation to the perturbed singular value:

$$\tilde{\sigma}_i \overset{1}{=} \sigma_i + u_i^T \Delta A v_i, \tag{1.1}$$

where u_i is the left singular vector of the matrix A corresponding to the singular value σ_i.

Stewart [15] provides a counterexample showing that, although $\|\Delta A\|$ may be small, the change in the angle of the singular vector can still be large. This implies that $\|\Delta v_i\|$ is not necessarily controlled by $\|\Delta A\|$ alone. Suppose A is a symmetric matrix such that $Av_i = \sigma_i v_i$. According to Theorem 6.1 of Jia and Stewart [8], we obtain an upper bound for $\sin\theta$ between v_i and $\tilde{v}_i$:

$$\sin\angle(v_i, \tilde{v}_i) \leq \frac{\|\Delta A\|}{\min\limits_{j\neq i} |\tilde{\sigma}_i - \sigma_j|} \triangleq \frac{\|\Delta A\|}{\delta}, \tag{1.2}$$

where σ_j is the j-th largest singular value of the matrix A.

Suppose $\|\Delta A\|$ is sufficiently small so that $\tilde{\sigma}_i$ remains close to σ_i. When the matrix A has a multiple singular value at σ_i, or has other singular values that are very close to σ_i, this causes the spectral gap $\delta \to 0^+$. In this case, according to inequality (1.2), the angle between the original singular vector v_i and its perturbed counterpart $\tilde{v}_i$ is not necessarily small. This explains why the example in the introduction of Stewart [15] serves as a counterexample. Therefore, one cannot always approximate the perturbed singular vector using a Taylor-like expansion alone, especially in the case of multiple singular values or when the singular values are closely cluster.

Although this study does not directly employ the *sin* Θ theory and indeed applies some Taylor-like expansion implicitly, it avoids applying such expansions directly to individual perturbed singular vectors, as this can become unstable in the case of repeated or nearly repeated singular values. And to preserve the structure of the SVD, we introduce an orthogonal alignment step based on the *Orthogonal Procrustes problem* (see Schönemann [14]). To the best of our knowledge, this is the first study to apply the orthogonal Procrustes approach to the perturbation of SVD. A detailed discussion of this methodology can be found in Chapter 3.

Chapter 2

Low-Rank Modification of SVD

Suppose we have an SVD of the matrix $A \in \mathbb{R}^{n\times m}$

$$A = \begin{bmatrix} U & U_\perp \end{bmatrix} \begin{bmatrix} \Sigma & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} V^T \\ V_\perp^T \end{bmatrix} = U\Sigma V^T, \tag{2.1}$$

where $U \in \mathbb{R}^{n\times r}$ and $V \in \mathbb{R}^{m\times r}$ are orthogonal matrices, $\Sigma \in \mathbb{R}^{r\times r} = \mathrm{diag}(\sigma_1, ..., \sigma_r)$ is a diagonal matrix with $\sigma_1 \geq ... \geq \sigma_r > 0$, $r = rank(A)$, $U_\perp \in \mathbb{R}^{n\times(n-r)}$ and $V_\perp \in \mathbb{R}^{m\times(m-r)}$ are orthogonal matrices satisfying $U^T U_\perp = \mathbf{0}$ and $V^T V_\perp = \mathbf{0}$.

We aim to derive, rather than directly evaluate, an SVD of the updated matrix $\hat{A} := A+E$, where $rank(E) = l \leq O(\sqrt{n \wedge m})$. This is a low-rank modification problem. Since $rank(E) = l$, $E = CD^T$, where $C \in \mathbb{R}^{n\times l}$ and $D \in \mathbb{R}^{m\times l}$.

First, we follow the approach by Brand [2] to reduce the original problem—evaluating the SVD of the modified matrix $\hat{A}$—to the problem of evaluating the SVD of a low-rank modification of a diagonal matrix. We express $\hat{A}$ as the product of three block matrices, as shown in equation (2.1).

$$\hat{A} = A + E = U\Sigma V^T + CD^T = \begin{bmatrix} U & C \end{bmatrix} \begin{bmatrix} \Sigma & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} V^T \\ D^T \end{bmatrix}. \tag{2.2}$$

Suppose that the square root matrix $R_C := (C^T U_\perp U_\perp^T C)^{\frac{1}{2}} \in \mathbb{R}^{l\times l}$ is well-defined, then we let $P := U_\perp U_\perp^T C (C^T U_\perp U_\perp^T C)^{\frac{-1}{2}}$. This leads the following equation

$$\begin{bmatrix} U & C \end{bmatrix} = \begin{bmatrix} U & P \end{bmatrix} \begin{bmatrix} I & U^T C \\ \mathbf{0} & R_C \end{bmatrix}, \tag{2.3}$$

where the matrix $\begin{bmatrix} U & P \end{bmatrix}$ is an orthogonal matrix. The equation (2.3) represents the projection of the column vectors of C onto the subspaces spanned by U and $U_\perp$ respectively. Similarly to (2.3), for the matrix $\begin{bmatrix} V & D \end{bmatrix}$, we have

$$\begin{bmatrix} V & D \end{bmatrix} = \begin{bmatrix} V & Q \end{bmatrix} \begin{bmatrix} I & V^T D \\ \mathbf{0} & R_D \end{bmatrix}, \tag{2.4}$$

where $R_D := (D^T V_\perp V_\perp^T D)^{\frac{1}{2}}$ and $Q := V_\perp V_\perp^T D (D^T V_\perp V_\perp^T D)^{\frac{-1}{2}}$. Combining equations (2.2), (2.3), and (2.4), we obtain

$$\begin{aligned} \hat{A} = A + E &= \begin{bmatrix} U & P \end{bmatrix} \begin{bmatrix} I & U^T C \\ \mathbf{0} & R_C \end{bmatrix} \begin{bmatrix} \Sigma & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} I & V^T D \\ \mathbf{0} & R_D \end{bmatrix}^T \begin{bmatrix} V^T \\ Q^T \end{bmatrix} \\ &\triangleq \begin{bmatrix} U & P \end{bmatrix} K \begin{bmatrix} V^T \\ Q^T \end{bmatrix}. \end{aligned} \tag{2.5}$$

The matrices $\begin{bmatrix} U & P \end{bmatrix}$ and $\begin{bmatrix} V^T \\ Q^T \end{bmatrix}$ are orthogonal matrices, thus the remaining work is to derive an SVD of the matrix K which is defined implicitly in the above equation. It is worth noting that the matrix K can be represented as the sum of a diagonal matrix and l rank-one matrices (refer to (2.6)). This implies that the problem of deriving the SVD of a low-rank modified matrix can be reduced to analyzing a diagonal matrix updated by a low-rank matrix of the same rank.

$$\begin{aligned} K &:= \begin{bmatrix} I & U^T C \\ \mathbf{0} & R_C \end{bmatrix} \begin{bmatrix} \Sigma & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} I & V^T D \\ \mathbf{0} & R_D \end{bmatrix}^T \\ &= \begin{bmatrix} \Sigma & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} + \begin{bmatrix} U^T C \\ R_C \end{bmatrix} \begin{bmatrix} V^T D \\ R_D \end{bmatrix}^T \\ &= \begin{bmatrix} \Sigma & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} + \sum_{i=1}^{l} \hat{C}_i \hat{D}_i^T, \end{aligned} \tag{2.6}$$

where $\begin{bmatrix} U^T C \\ R_C \end{bmatrix} \triangleq \hat{C} \triangleq \begin{bmatrix} \hat{C}_1 \cdots \hat{C}_l \end{bmatrix} \in \mathbb{R}^{(r+l)\times l}$ and $\begin{bmatrix} V^T D \\ R_D \end{bmatrix} \triangleq \hat{D} \triangleq \begin{bmatrix} \hat{D}_1 \cdots \hat{D}_l \end{bmatrix} \in \mathbb{R}^{(r+l)\times l}$, each $\hat{C}_i$ and $\hat{D}_i$ is a column vector.

Observe that the matrix $K \in \mathbb{R}^{(r+l)\times(r+l)}$, where $r = rank(A)$ and $l = rank(E) \leq O(\sqrt{n \wedge m})$. If the matrix A is low-rank, precisely, $r = rank(A) \leq O(\sqrt{n \wedge m})$, then the

size of the matrix K is small. Consequently, computing an SVD of the matrix K directly is not computationally expensive, and then by (2.5), an SVD of the updated matrix $\hat{A}$ is done efficiently.

However, when the matrix A is not low-rank, the size of the matrix K may become too large for computing its SVD directly in an efficient manner. In this study, motivated by the structure presented in equation (2.6), we adopt a strategy that iteratively applies the rank-one SVD update l times to compute the SVD of the matrix K, as illustrated in Figure 2.1.

$$K = \underbrace{\underbrace{\underbrace{\begin{bmatrix} \Sigma & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} + \hat{C}_1 \hat{D}_1^T}_{\text{1st rank-one update}} + \hat{C}_2 \hat{D}_2^T}_{\text{2nd rank-one update}} + \cdots + \hat{C}_l \hat{D}_l^T}_{l\text{-th rank-one update}}$$

Figure 2.1: Successive rank-one updates applied to the matrix K, where $l \leq O(\sqrt{n \wedge m})$.

Based on the previous derivation, each rank-one update can be reduced to a rank-one modification of a diagonal matrix. Specifically, the key technical problem then lies in computing the SVD of a diagonal matrix plus a rank-one matrix which forms as $\Lambda + uv^T$. We will elaborate on the procedure for deriving the SVD of such a matrix in the following section.

2.1 Rank-One Modification of a Diagonal Matrix

Let $t \in \mathbb{N}$. Let $\Lambda \in \mathbb{R}^{t \times t}$ be a diagonal matrix. Let u and v be two vectors in $\mathbb{R}^t$. Define $C := \Lambda + uv^T$. We need to evaluate an SVD of the matrix C. In this section, we organize the results of Gandhi et al. [5], Golub [6], and Wilkinson [20] related to the SVD of a rank-one modification of a diagonal matrix.

Consider

$$C^T C = \Lambda^2 + \|u\|^2 vv^T + \hat{u}v^T + v\hat{u}^T, \tag{2.7}$$

where $\hat{u} \triangleq \Lambda u$ and $\hat{v} \triangleq \Lambda v$.

Then we have

$$
\begin{aligned}
C^TC &= \Lambda^2 + (\|u\|^2 v + \hat{u})v^T + v\hat{u}^T \\
&= \Lambda^2 + \begin{bmatrix} \|u\|^2 v + \hat{u} & v \end{bmatrix} \begin{bmatrix} v^T \\ \hat{u}^T \end{bmatrix} \\
&= \Lambda^2 + \begin{bmatrix} v & \hat{u} \end{bmatrix} \begin{bmatrix} \|u\|^2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v^T \\ \hat{u}^T \end{bmatrix}.
\end{aligned} \tag{2.8}
$$

Since $\begin{bmatrix} \|u\|^2 & 1 \\ 1 & 0 \end{bmatrix}$ is real and symmetric, $\begin{bmatrix} \|u\|^2 & 1 \\ 1 & 0 \end{bmatrix} = Q \begin{bmatrix} \rho_1 & 0 \\ 0 & \rho_2 \end{bmatrix} Q^T$, where $Q \in \mathbb{R}^{2\times 2}$ is an unitary. Denote $\begin{bmatrix} v & \hat{u} \end{bmatrix} Q \triangleq \begin{bmatrix} \tilde{v} & \tilde{u} \end{bmatrix}$, then we have

$$
C^TC = \Lambda^2 + \rho_1 \tilde{v}\tilde{v}^T + \rho_2 \tilde{u}\tilde{u}^T. \tag{2.9}
$$

We would present the method for evaluating the orthogonal diagonalization of symmetric form $D + \rho ww^T$ in Subsection 2.1.1, where D is a diagonal matrix, ρ is a real number, and w is a vector. For now, we assume that this procedure is known.

Suppose we have $\Lambda^2 + \rho_1 \tilde{v}\tilde{v}^T = Q_1 \Lambda_1 Q_1^T$, where $Q_1 \in \mathbb{R}^{t\times t}$ is an unitary and Λ_1 is a diagonal matrix. And denote $y \triangleq Q_1^T \tilde{u}$, then we obtain

$$
C^TC = Q_1(\Lambda_1 + \rho_2 yy^T)Q_1^T. \tag{2.10}
$$

Again, assume we have an orthogonal diagonalization $\Lambda_1 + \rho_2 yy^T = Q_2 \Lambda_2 Q_2^T$, then

$$
C^TC = (Q_1Q_2)\Lambda_2(Q_1Q_2)^T. \tag{2.11}
$$

The equation (2.11) gives an orthogonal diagonalization of the symmetric matrix C^TC. Then it is straightforward to obtain an SVD of the matrix C. Thus the remaining work is to derive an algorithm to compute an orthogonal diagonalization of symmetric type rank-one update $D + \rho ww^T$.

2.1.1 Orthogonal Diagonalization of the Matrix $D + \rho ww^T$

Let $D = \mathrm{diag}(d_1, \cdots, d_t)$ with $d_1 \geq \cdots \geq d_t$. Let $w \in \mathbb{R}^t$ and let $\rho \in \mathbb{R}$. Define $C := D + \rho ww^T$. Thus, the matrix C is symmetric and therefore orthogonally diagonalizable. Let $\lambda_1 \geq \cdots \geq \lambda_t$ be the eigenvalues of the matrix C. Wilkinson [20] has provided precise bounds for each eigenvalue of the matrix C:

(i) if $\rho > 0$, then $\begin{cases} d_1 \le \lambda_1 \le d_1 + \rho\|w\|^2 \\ d_i \le \lambda_i \le d_{i-1}, \quad i = 2, \cdots, t; \end{cases}$

(ii) if $\rho < 0$, then $\begin{cases} d_t + \rho\|w\|^2 \le \lambda_t \le d_t \\ d_{i+1} \le \lambda_i \le d_i, \quad i = 1, \cdots, t-1. \end{cases}$

In order to evaluate the exact eigenvalues of the matrix C, we turn to solving the characteristic polynomial equation

$$\det(C - xI) = 0. \tag{2.12}$$

Suppose $x \notin \{d_1, \cdots, d_t\}$, then $D - xI$ is invertible. And it is straightforward to verify the following equation

$$\det(C - xI) = \det(D - xI)\ \det\big(I + \rho(D - xI)^{-1} ww^T\big). \tag{2.13}$$

Since $D - xI$ is invertible, $\det(D - xI) \neq 0$, then it is suffices to solve

$$\det\big(I + \rho(D - xI)^{-1} ww^T\big) = 0. \tag{2.14}$$

The following equation is an immediate consequence of the *Matrix determinant lemma* (see Ding et al. [3]). We give a proof of its reduced form in Appendix A.1.

$$\det\big(I + \rho(D - xI)^{-1} ww^T\big) = 1 + \rho w^T (D - xI)^{-1} w. \tag{2.15}$$

Denote $w \triangleq \begin{bmatrix} w_1 & \cdots & w_t \end{bmatrix}^T \in \mathbb{R}^t$. A direct computing yields

$$1 + \rho w^T (D - xI)^{-1} w = 1 + \rho \sum_{i=1}^{t} \frac{w_i^2}{d_i - x}. \tag{2.16}$$

Therefore, the eigenvalues of the matrix C can be evaluated by finding the roots of the following equation

$$f(x) := 1 + \rho \sum_{i=1}^{t} \frac{w_i^2}{d_i - x} = 0. \tag{2.17}$$

The domain of the real-valued function f is $\mathbb{R} \setminus \{d_1, \cdots, d_t\}$.

Note that $f'(x) = \rho \sum_{i=1}^{t} \frac{w_i^2}{(d_i - x)^2}$ for x in the domain of the function f. This implies that $f' \ge 0$ if $\rho > 0$; otherwise, $f' \le 0$. The monotonicity of the function f, together with its graphical behavior (see Figure 2.2), suggests the use of the bisection method or Newton's method on each interval provided by Wilkinson [20] to evaluate each root of f. In practice, Newton's method is commonly employed to compute the t roots, with a total computational

complexity of $O\Big(t^2 \log_2\big(\log_2(1/\epsilon)\big)\Big)$, where ϵ represents the machine precision. And it should be noted that the initial guess for Newton's method should not be too close to any d_i, as $f(x)$ has vertical asymptotes at $x = d_i$ and the derivative $f'(x)$ may become large in magnitude near these points, potentially causing divergence.

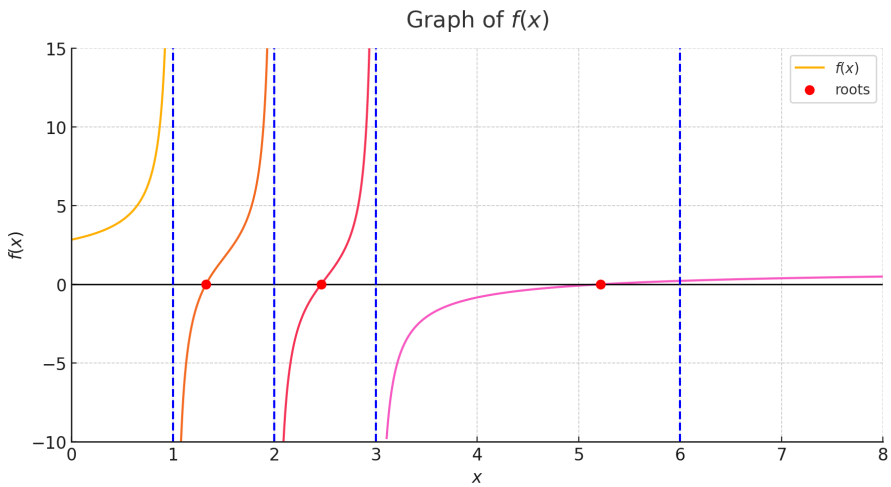

Figure 2.2: Graph of the function $f(x)$ with parameters $\rho = 1$, $t = 3$, $w_1 = w_2 = w_3 = 1$, and $D = \mathrm{diag}(3, 2, 1)$. According to Wilkinson [20], in this case, Newton' s method is applied on the intervals $[1, 2]$, $[2, 3]$, and $[3, 6]$ to locate the roots. The roots of the function f are $\lambda_1 \approx 5.21$, $\lambda_2 \approx 2.46$, and $\lambda_3 \approx 1.33$ (rounded to two decimal places).

Suppose that the eigenvalues $\lambda_1, \ldots, \lambda_t$ of the matrix $C = D + \rho w w^T$ are already known. Then, we proceed to determine the eigenvectors associated with these eigenvalues. Therefore, we solve

$$Cx_i = \lambda_i x_i, \quad i = 1, \cdots, t, \tag{2.18}$$

for each eigenvector x_i corresponding to eigenvalue λ_i of the matrix C. Suppose $D - \lambda_i I$ is invertible, then we have

$$x_i = (-\rho w^T x_i)(D - \lambda_i I)^{-1} w, \quad i = 1, \cdots, t. \tag{2.19}$$

Take

$$\hat{x}_i = \frac{(D - \lambda_i I)^{-1} w}{\|(D - \lambda_i I)^{-1} w\|}, \quad i = 1, \cdots, t. \tag{2.20}$$

Then $\hat{x}_i$ is an unit vector and an eigenvector of C corresponding to the eigenvalue λ_i. Since the matrix C is real and symmetric, $\hat{x}_i$ is perpendicular to $\hat{x}_j$ if $\lambda_i \neq \lambda_j$. Define $\hat{X} := \begin{bmatrix} \hat{x}_1 & \cdots & \hat{x}_t \end{bmatrix} \in \mathbb{R}^{t \times t}$. Then $\hat{X}$ is an unitary. Finally, we derive an orthogonal diagonalization of the matrix C

$$C = \hat{X} \mathrm{diag}(\lambda_1, \cdots, \lambda_t) \hat{X}^T. \tag{2.21}$$

It should be noted that, strictly speaking, the above equation does not hold exactly, since the values of λ_i obtained via Newton's method may contain slight numerical errors (albeit possibly within machine precision). Consequently, these inaccuracies may propagate to the computed eigenvectors $\hat{x}_i$, introducing small numerical errors as well.

2.1.2 A Brief Discussion on the Computational Cost of Rank-One Updates

Notice that equation (2.11) involves a dense matrix-matrix multiplication Q_1Q_2, where both of Q_1 and Q_2 can be factorized as a diagonal matrix, followed by a *Cauchy matrix* (whose (i,j)-th entry takes the form $\frac{1}{\sigma_i-\lambda_j}$) , and another diagonal matrix. The product Q_1Q_2 requires approximately $(r+l)^3$ operations by the natural definition of matrix-matrix multiplication ($r+l$ may not be a small number as A is not low-rank). Consequently, repeating this computation over l iterations (cf. Figure 2.1) results in a total cost of approximately $l(r+l)^3$ operations. Such expensive computational cost makes our algorithm difficult to implement in practice.

Gandhi et al. [5] present an efficient way to evaluate the matrix product Q_1Q_2. To reduce the computational expense, they employ the *Fast Multipole Method (FMM)*. This gain in efficiency, however, comes at the expense of introducing approximation error. Gandhi et al. [5] claim that they proposed an efficient method that performs each rank-one SVD update in $O\big(n^2\log(\frac{1}{\epsilon})\big)$ operations, where n is the length of the vector and ϵ denotes the machine precision.

In addition to the work of Gandhi et al. [5], Baglama et al. [1] claim that their algorithm requires only quadratic complexity for the rank-one modification $C := \Lambda + uv^T$. Rather than converting the nonsymmetric form $\Lambda + uv^T$ into the symmetric type $D + \rho ww^T$ and then addressing that case, they directly analyze the nonsymmetric type $C = \Lambda + uv^T$. Baglama et al. [1] start from analyzing the characteristic polynomial of the matrix C^TC for singular values of the matrix C, and then proceed to solve for the corresponding singular vectors. Although the overall outline of Baglama et al.'s work [1] is similar to Subsection 2.1.1 of this thesis, their detailed derivation is considerably more involved such that the details of their work is beyond the scope of this thesis.

Chapter 3

Perturbation of the SVD

Let $A \in \mathbf{R}^{n\times m}$ be a real matrix. Suppose we have an SVD of the matrix A

$$A = \begin{bmatrix} U & U_\perp \end{bmatrix} \begin{bmatrix} \Sigma & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} V^T \\ V_\perp^T \end{bmatrix} = U\Sigma V^T, \tag{3.1}$$

where $U \in \mathbf{R}^{n\times p}$ and $V \in \mathbf{R}^{m\times p}$ are orthogonal matrices, $\Sigma \in \mathbf{R}^{p\times p} = \mathrm{diag}(\sigma_1, ..., \sigma_p)$ is a diagonal matrix with $\sigma_1 \geq ... \geq \sigma_p > 0$, $p = rank(A)$, $U_\perp \in \mathbf{R}^{n\times(n-p)}$ and $V_\perp \in \mathbf{R}^{m\times(m-p)}$ are orthogonal matrices satisfying $U^T U_\perp = \mathbf{0}$ and $V^T V_\perp = \mathbf{0}$.

Let ΔA be a perturbation of the matrix A. Suppose that there exists $\epsilon > 0$ such that $\|\Delta A\| \leq \epsilon < \sigma_p$, where σ_p is the smallest nonzero singular value of the matrix A. Define $\tilde{A} := A + \Delta A$. Then the matrix ΔA alters the original SVD structure of the matrix A. Our ultimate goal is to recover the SVD of the perturbed matrix $\tilde{A}$.

Consider

$$\tilde{A} \begin{bmatrix} V & V_\perp \end{bmatrix} = \begin{bmatrix} U\Sigma + \Delta A \cdot V & \Delta A \cdot V_\perp \end{bmatrix}. \tag{3.2}$$

Denote $U = \begin{bmatrix} u_1 & \cdots & u_p \end{bmatrix}$, $V = \begin{bmatrix} v_1 & \cdots & v_p \end{bmatrix}$, and $V_\perp = \begin{bmatrix} v_{p+1} & \cdots & v_m \end{bmatrix}$. Then we have

$$\tilde{A} \begin{bmatrix} V & V_\perp \end{bmatrix} = \begin{bmatrix} \sigma_1 u_1 + \Delta A \cdot v_1 & \cdots & \sigma_p u_p + \Delta A \cdot v_p & \Delta A \cdot v_{p+1} & \cdots & \Delta A \cdot v_m \end{bmatrix}. \tag{3.3}$$

The column vectors of the matrix $\tilde{A}V_\perp$ are dominated by the perturbation ΔA, and thus may exhibit poor orthogonality. In fact, these vectors can even become linearly dependent. This makes it more complicated to discuss SVD-preserving algorithms involving these vectors. Therefore, the rest of this study focuses solely on structure-preserving algorithms for the matrix $\tilde{A}V$.

Define $W := \tilde{A}V$ and denote $W = \begin{bmatrix} w_1 & \cdots & w_p \end{bmatrix}$. Since $\|\Delta A\| < \sigma_p$ and each u_i and v_i are unit vectors, it follows that w_i is not a zero vector for all i. Therefore, $\|w_i\| > 0, \quad \forall i$. Then

$$\begin{aligned} W &= \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} \operatorname{diag}(\|w_1\|, \cdots, \|w_p\|) \\ &\triangleq \hat{W} \operatorname{diag}(\|w_1\|, \cdots, \|w_p\|). \end{aligned} \tag{3.4}$$

The column vectors of the matrix $\hat{W}$ are unit vectors; however, their orthogonality is perturbed by the matrix ΔA. To recover the orthogonality, we normalize the matrix $\hat{W}$ by right-multiplying it with the matrix $\hat{N} := (\hat{W}^T\hat{W})^{\frac{-1}{2}} \in \mathbb{R}^{p\times p}$, referred to as the normalizer. Then we obtain the following equation

$$\tilde{A}V \operatorname{diag}(\|w_1\|, \cdots, \|w_p\|)^{-1}\hat{N} = \hat{W}\hat{N}. \tag{3.5}$$

$\hat{W}\hat{N}$ and V are orthogonal matrices. To gain insight into the behavior of the matrix $\operatorname{diag}(\|w_1\|, \cdots, \|w_p\|)^{-1}\hat{N}$, it is essential to investigate the structure of the normalizer $\hat{N}$. By the definition of $\hat{N} := (\hat{W}^T\hat{W})^{\frac{-1}{2}}$, this requires analyzing the properties of the matrix $\hat{W}^T\hat{W}$. Recall that each column vector of the matrix $\hat{W}$ has length 1, this implies $(\hat{W}^T\hat{W})_{i,i} = 1$ for each i. The proof of $\left|(\hat{W}^T\hat{W})_{i,j}\right| \leq O(\|\Delta A\|)$ for $i \neq j$ is provided in Appendix A.2. Then $\|\hat{W}^T\hat{W} - I\| \leq O(\|\Delta A\|)$. When $\|\Delta A\| \to 0^+$, we have the following approximation

$$\begin{aligned} \hat{N} &:= (\hat{W}^T\hat{W})^{\frac{-1}{2}} \\ &= \left[I + (\hat{W}^T\hat{W} - I)\right]^{\frac{-1}{2}} \\ &\overset{1}{=} I - \frac{1}{2}(\hat{W}^T\hat{W} - I) \\ &\overset{0}{=} I. \end{aligned} \tag{3.6}$$

In (3.6), the first approximation follows from the *Neumann expansion* (see Vaccaro [17]) and provides a first-order approximation of the matrix $\hat{N}$. And since $\|\hat{W}^T\hat{W} - I\| \leq O(\|\Delta A\|)$, the second approximation is a zeroth-order approximation of the normalizer $\hat{N}$. Therefore, as $\|\Delta A\| \to 0^+$, the matrix $\operatorname{diag}(\|w_1\|, \cdots, \|w_p\|)^{-1}\hat{N}$ approaches to the diagonal matrix $\operatorname{diag}(\|w_1\|, \cdots, \|w_p\|)^{-1}$.

However, it isn't a wise decision to simply replace the matrix $\operatorname{diag}(\|w_1\|, \cdots, \|w_p\|)^{-1}\hat{N}$ with the diagonal matrix $\operatorname{diag}(\|w_1\|, \cdots, \|w_p\|)^{-1}$ in the equation (3.5). The first reason is that the zeroth-order approximation $\hat{N} \overset{0}{=} I$ is not sufficiently accurate. Secondly, the right singular vectors V remain unmodified (cf. (3.5)) if $\hat{N}$ is replaced by the identity matrix I, which fails to achieve our intended purpose.

Nevertheless, the approximation $\mathrm{diag}(\|w_1\|, \cdots, \|w_p\|)^{-1}\hat{N} \overset{0}{=} \mathrm{diag}(\|w_1\|, \cdots, \|w_p\|)^{-1}$ inspires a crucial idea that guides our subsequent development. Note that any diagonal matrix can be expressed as a matrix product $Q\Lambda$, where Q is an unitary and Λ is a diagonal matrix. Then we turn to find an unitary Q and a diagonal matrix Λ to approximate the matrix $\mathrm{diag}(\|w_1\|, \cdots, \|w_p\|)^{-1}\hat{N}$ as closely as possible. This consideration leads us to formulate the following minimization problem

$$\min_{\substack{Q \text{ unitary} \\ \Lambda \text{ diagonal matrix}}} \left\| \mathrm{diag}(\|w_1\|, \cdots, \|w_p\|)^{-1}\hat{N} - Q\Lambda \right\|_F. \tag{3.7}$$

Then such unitary Q and diagonal matrix Λ transform the equation (3.5) into the following approximation

$$\tilde{A}VQ\Lambda \approx \hat{W}\hat{N}. \tag{3.8}$$

Both of V and Q are orthogonal matrix, so does the product VQ. The above approximation offers two main advantages: (1) it preserves the SVD structure, and (2) it updates the right singular vectors V as the orthogonal matrix VQ. A detailed discussion on whether the residual $\|\tilde{A}VQ\Lambda - \hat{W}\hat{N}\|$ tends to zero as $\|\Delta A\| \to 0^+$ will be given in Section 3.1.

The remaining problem is how to find an appropriate unitary Q and a diagonal matrix Λ in the minimization problem (3.7). Define $\hat{S} := \mathrm{diag}(\|w_1\|, \cdots, \|w_p\|)^{-1}\hat{N}$ and denote $\hat{S} = \begin{bmatrix} \hat{s}_1 & \cdots & \hat{s}_p \end{bmatrix}$. In (3.7), since Q should be an unitary, each column vector of the matrix Q is a unit vetor. This suggests us to take Λ as $\mathrm{diag}(\|\hat{s}_1\|, \cdots, \|\hat{s}_p\|)$. Then the minimization problem (3.7) is reduced to the optimization problem below

$$\min_{Q \text{ unitary}} \left\| \hat{S} - Q\Lambda \right\|_F^2. \tag{3.9}$$

It should be emphasized that the minimization problems (3.7) and (3.9) are not equivalent problem. The diagonal matrix $\Lambda := \mathrm{diag}(\|\hat{s}_1\|, \cdots, \|\hat{s}_p\|)$ may not be the solution to the problem (3.7). Despite this, one may reasonably expect that the solutions to (3.7) and (3.9) are close.

The reduced minimization problem (3.9) is a typical instance of the *orthogonal Procrustes problem* (refer to Schönemann [14]). Suppose that $U'\Sigma'(V')^T$ is an SVD of the matrix $\hat{S}\Lambda$, then one can show that the solution in (3.9) is $Q = U'(V')^T$ (see Appendix A.3 for the proof). Consequently, we obtain the following approximation, which is the main result of this thesis

$$\tilde{A}VU'(V')^T\Lambda \approx \hat{W}\hat{N}. \tag{3.10}$$

Before proceeding to Section 3.1, we use the following table to summarize the computational expense required for the SVD approximation in (3.10), based on the sequence of operations involved in its computation:

Computation Order	Operation	Computational Cost
1	Matrix-matrix multiplication $W := \tilde{A}V$	$\approx nmp$
2	Normalize W to derive $\hat{W}$	$\approx np$
3	Matrix-matrix multiplication $\hat{W}^T\hat{W}$	$\approx np^2$
4	Evaluate $\hat{N} := (\hat{W}^T\hat{W})^{\frac{-1}{2}}$	$O(p^3)$
5	Matrix-matrix multiplication $\hat{W}\hat{N}$	$\approx np^2$
6	Evaluate Λ	$\approx p^2$
7	Evaluate $\hat{S}$	$\approx p^2$
8	Matrix-matrix multiplication $\hat{S}\Lambda$	$\approx p^2$
9	Compute an SVD of $\hat{S}\Lambda = U'\Sigma'(V')^T$	$O(p^3)$
10	Matrix-matrix multiplication $U'(V')^T$	$\approx p^3$
11	Matrix-matrix multiplication $V[U'(V')^T]$	$\approx mp^2$

Table 3.1: Computational steps and associated computational expense of each operation in the SVD approximation (3.10). This table shows that: (1) the modified left singular vectors $\hat{W}\hat{N}$ require approximately $nmp+O(np^2)$ operations, (2) the modified singular values Λ require $O(p^2)$ operations, and (3) the modified right singular vectors $V[U'(V')^T]$ require $O(mp^2)$ operations. Therefore, compared to directly computing the SVD of the perturbed matrix $\tilde{A}\big|_{N(A)^\perp}$, the approximation in (3.10) incurs lower computational cost.

3.1 Error Analysis

In this section, we focus on analyzing the convergence of the residual associated with the approximation in (3.10):

$$\left\|\tilde{A}VU'(V')^T\Lambda - \hat{W}\hat{N}\right\|. \tag{3.11}$$

It is easy to verify the following equation

$$\left\|\tilde{A}VU'(V')^T\Lambda - \hat{W}\hat{N}\right\| = \left\|\tilde{A}VU'\left[(V')^T\Lambda - \Sigma'(V')^T\Lambda^{-1}\right]\right\|. \tag{3.12}$$

According to (3.12), to study the convergence of the residual $\left\|\tilde{A}VU'(V')^T\Lambda - \hat{W}\hat{N}\right\|$, it suffices to analyze the behavior of $\|(V')^T\Lambda - \Sigma'(V')^T\Lambda^{-1}\|$ (cf. Liu et al. [10]). Specifically, we aim to

show that

$$\left\|(V')^T\Lambda - \Sigma'(V')^T\Lambda^{-1}\right\| \to 0^+ \text{ as } \|\Delta A\| \to 0^+. \tag{3.13}$$

Denote $\Sigma' \triangleq \operatorname{diag}(\sigma'_1, \cdots, \sigma'_p)$. A direct evaluation gives

$$\left((V')^T\Lambda - \Sigma'(V')^T\Lambda^{-1}\right)_{i,j} = \left(\|\hat{s}_j\| - \frac{\sigma'_i}{\|\hat{s}_j\|}\right)(V')_{j,i}, \quad i,j \in \{1, \cdots, p\}. \tag{3.14}$$

And one can show that (see Appendix A.4 for the proof)

$$\sigma'_i = \left[\sum_{j=1}^{p} \|\hat{s}_j\|^4 \left((V')_{j,i}\right)^2\right]^{\frac{1}{2}}, \quad i = 1, \cdots, p. \tag{3.15}$$

In addition, recall that

$$\begin{aligned} \hat{S}\Lambda &= U'\Sigma'(V')^T \\ &= \operatorname{diag}(\|w_1\|, \cdots, \|w_p\|)^{-1}\left[I + \left(\hat{W}^T\hat{W} - I\right)\right]^{\frac{-1}{2}} \operatorname{diag}(\|\hat{s}_1\|, \cdots, \|\hat{s}_p\|). \end{aligned} \tag{3.16}$$

A rigorous proof of convergence in (3.13) is beyond the scope of our current analysis. Nevertheless, we can provide an informal justification to support the plausibility of this convergence. Recall that $\|\hat{W}^T\hat{W} - I\| \leq O(\|\Delta A\|)$. As $\|\Delta A\| \to 0^+$, $\left[I + \left(\hat{W}^T\hat{W} - I\right)\right]^{\frac{-1}{2}} \overset{0}{=} I$ by (3.6). This implies that the matrix $\hat{S}\Lambda$ is almost a diagonal matrix by (3.16), then this may lead the matrix V' approaches to the identity matrix. In other words,

$$(V')_{j,i} \approx \begin{cases} 0, & \text{if } j \neq i, \\ 1, & \text{if } j = i. \end{cases} \tag{3.17}$$

Combine (3.14) and (3.17), we have $\left((V')^T\Lambda - \Sigma'(V')^T\Lambda^{-1}\right)_{i,j} \approx 0$ when $i \neq j$. As for the case of $i = j$, applying (3.14), (3.15), and (3.17) gives $\left((V')^T\Lambda - \Sigma'(V')^T\Lambda^{-1}\right)_{i,i} \approx 0$. This argument informally suggests that the residual $\left\|\tilde{A}VU'(V')^T\Lambda - \hat{W}\hat{N}\right\| \to 0^+$ as $\|\Delta A\| \to 0^+$.

3.1.1 Numerical Justification

Due to the lack of a rigorous proof for the convergence of the residual, we resort to numerical results to support the claim of convergence. The design of the following numerical experiment is inspired by the numerical simulations presented in the work of Vaccaro [17].

Let

$$A = \begin{bmatrix} 2.5 & -0.7 & 3.2 & 7.6 & -1.2 & -1.2 & 7.9 & 3.8 & -2.3 & 2.7 \\ -2.3 & -2.3 & 1.2 & -9.6 & -8.6 & -2.8 & -5.1 & 1.6 & -4.5 & -7.1 \\ 7.3 & -1.1 & 0.3 & -7.1 & -2.7 & 0.6 & -5.8 & 1.9 & -3.0 & -1.5 \\ -3.0 & 9.3 & -0.1 & -5.3 & 4.1 & -6.1 & 1.0 & -9.8 & -6.6 & 1.0 \\ 3.7 & 0.9 & -0.6 & -1.5 & -7.4 & -3.6 & -2.3 & 5.3 & 1.7 & -8.8 \\ 1.6 & -1.9 & -3.4 & 3.1 & 5.2 & 4.7 & -4.2 & -1.5 & 1.7 & 4.9 \\ -2.4 & -0.9 & -5.5 & -6.0 & 4.1 & 6.8 & -0.4 & 5.0 & 1.8 & -3.2 \\ 1.8 & 7.7 & -0.2 & 7.8 & -13.1 & 4.1 & 0.4 & -1.5 & 0.5 & -9.9 \\ -1.1 & 1.8 & 7.4 & -2.6 & -4.0 & -2.5 & 4.6 & 1.6 & -2.6 & 2.6 \\ 0.5 & 4.8 & -3.5 & -1.6 & -2.0 & -7.3 & 1.5 & 1.3 & 0.0 & -1.2 \end{bmatrix}. \tag{3.18}$$

Each entry of the matrix $A \in \mathbb{R}^{10\times 10}$ is drawn from a normal distribution with mean 0 and standard deviation 5. All entries are then rounded to the first decimal place. The reason for using randomized matrices in our numerical experiment is to demonstrate that the residual associated with (3.10) converges to zero across a variety of scenarios consistent with the condition $\|\Delta A\| \to 0^+$.

The smallest singular value of the matrix A in (3.18) is 1.14 (rounded to the second decimal place). As assumed in the earlier theoretical derivation, this requires that $\|\Delta A\| < 1.14$. To test our theory, we examine a sequence of perturbations where $\|\Delta A\|$ ranges from 10^{-12} to $10^0 = 1$, increasing by a factor of 10 at each step. For each value of $\|\Delta A\| = 10^i$ $(i = -12, -11, \ldots, 0)$, 10,000 realizations of the perturbed matrix $\tilde{A} = A + \Delta A$ were generated. The perturbation matrix ΔA in each realization was constructed by first drawing the entries of ΔA independently from the standard normal distribution, and then scaling the matrix such that $\|\Delta A\| = 10^i$. For each 10^i, the average residual over the 10,000 realizations was computed. The results are presented in the following table.

$\|\Delta A\|$		10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
Mean residual		3.569e-02	3.573e-03	3.564e-04	3.579e-05	3.571e-06	3.572e-07
$\|\Delta A\|$	10^{-6}	10^{-7}	10^{-8}	10^{-9}	10^{-10}	10^{-11}	10^{-12}
Mean residual	3.566e-08	3.567e-09	3.561e-10	3.562e-11	3.576e-12	3.556e-13	3.654e-14

Table 3.2: Mean residuals corresponding to different values of $\|\Delta A\|$, where the indicated residual is $\left\|\tilde{A}VU'(V')^T\Lambda - \hat{W}\hat{N}\right\|$. The data illustrated in this table suggest that residual may converge to zero with a convergence rate of $O(\|\Delta A\|)$ as $\|\Delta A\| \to 0^+$.

Chapter 4

Conclusions

Based on the works of Brand [2], Gandhi et al. [5], Golub [6], and Wilkinson [20], this study presents an algorithm for the low-rank modification of the SVD. However, how to balance the trade-off between computational efficiency and accuracy when deriving the SVD of a low-rank modified matrix $\hat{A}$ remains a question worthy of careful consideration.

In addition to addressing the low-rank modification of the SVD, this study places greater emphasis on the SVD perturbation problem. By employing an orthogonal Procrustes approach in conjunction with a Taylor-like expansion, we derive an SVD approximation for $\tilde{A}\big|_{N(A)^\perp}$ (cf. (3.10)). To the best of our knowledge, this represents one of the first systematic attempts to incorporate the orthogonal Procrustes problem into the analysis of SVD perturbations. In terms of results, approximation (3.10) perfectly preserves the structure of the SVD. As for the induced residual (cf. (3.11)), the results of the numerical experiments (cf. Subsection 3.1.1) suggests that the convergent rate is of order $O(\|\Delta A\|)$ when $\|\Delta A\| < \sigma_p$, where σ_p is the smallest nonzero singular value of the original matrix A. And the computational cost of the approximation in (3.10) is lower than that of directly evaluating the SVD of $\tilde{A}\big|_{N(A)^\perp}$ (cf. Table 3.1).

However, the proposed approach to the SVD perturbation problem has a notable limitation —it does not readily extend to analyzing the SVD of the component $\tilde{A}\big|_{N(A)}$ (cf. (3.3)), due to the uncertainty regarding the orthogonality of the matrix $\tilde{A}V_\perp$. As a consequence, this study does not offer a complete SVD approximation for the perturbed matrix $\tilde{A}$.

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Appendix

A.1 $\det(I + uv^T) = 1 + v^T u$

Claim. $\det(I + uv^T) = 1 + v^T u$, where $n \in \mathbb{N}$, $I \in \mathbb{R}^{n \times n}$, and $u, v \in \mathbb{R}^n$.

Proof. Ding et al. [3] presents the following identity, which can be readily verified

$$\begin{bmatrix} I & \mathbf{0} \\ v^T & 1 \end{bmatrix} \begin{bmatrix} I + uv^T & u \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} I & \mathbf{0} \\ -v^T & 1 \end{bmatrix} = \begin{bmatrix} I & u \\ \mathbf{0} & 1 + v^T u \end{bmatrix}. \tag{4.1}$$

Note that $\begin{bmatrix} I & \mathbf{0} \\ v^T & 1 \end{bmatrix}$ and $\begin{bmatrix} I & \mathbf{0} \\ -v^T & 1 \end{bmatrix}$ are lower triangular matrices, thus $\det(\begin{bmatrix} I & \mathbf{0} \\ v^T & 1 \end{bmatrix}) = \det(\begin{bmatrix} I & \mathbf{0} \\ -v^T & 1 \end{bmatrix}) = 1$. Similarly, $\begin{bmatrix} I & u \\ \mathbf{0} & 1 + v^T u \end{bmatrix}$ is an upper triangular matrix, then $\det(\begin{bmatrix} I & u \\ \mathbf{0} & 1 + v^T u \end{bmatrix}) = 1 + v^T u$. Therefore, we have

$$\det(\begin{bmatrix} I + uv^T & u \\ \mathbf{0} & 1 \end{bmatrix}) = 1 + v^T u. \tag{4.2}$$

Apply Laplace expansion to the left hand side of (4.2), we obtain

$$\det(I + uv^T) = 1 + v^T u. \tag{4.3}$$

□

A.2 Off-diagonal terms of $\hat{W}^T\hat{W}$

Claim. For $i \neq j$, we have

$$\left|(\hat{W}^T\hat{W})_{i,j}\right| \leq O(\|\Delta A\|).$$

Proof. Consider

$$\begin{aligned}
(\hat{W}^T\hat{W})_{i,j} &= \langle \hat{w}_i, \hat{w}_j \rangle = \langle \frac{w_i}{\|w_i\|}, \frac{w_j}{\|w_j\|} \rangle = \frac{1}{\|w_i\|\|w_j\|}\langle w_i, w_j \rangle \\
&= \frac{1}{\|w_i\|\|w_j\|}\langle \sigma_i u_i + \Delta A \cdot v_i \ , \ \sigma_j u_j + \Delta A \cdot v_j \rangle \\
&= \frac{1}{\|w_i\|\|w_j\|}\big(\langle \sigma_i u_i, \sigma_j u_j \rangle + \langle \sigma_i u_i, \Delta A \cdot v_j \rangle + \langle \Delta A \cdot v_i, \sigma_j u_j \rangle + \langle \Delta A \cdot v_i, \Delta A \cdot v_j \rangle\big) \\
&\overset{i \neq j}{=} \frac{1}{\|w_i\|\|w_j\|}\big(\langle \sigma_i u_i, \Delta A \cdot v_j \rangle + \langle \Delta A \cdot v_i, \sigma_j u_j \rangle + \langle \Delta A \cdot v_i, \Delta A \cdot v_j \rangle\big) \\
&= \frac{1}{\|w_i\|\|w_j\|}\big(\sigma_i \langle u_i, \Delta A \cdot v_j \rangle + \sigma_j \langle \Delta A \cdot v_i, u_j \rangle + \langle \Delta A \cdot v_i, \Delta A \cdot v_j \rangle\big).
\end{aligned} \tag{4.4}$$

We have assumed that $\|\Delta A\| \leq \epsilon < \sigma_p$. Thus

$$\begin{aligned}
\left|(\hat{W}^T\hat{W})_{i,j}\right| &\leq \frac{1}{\|w_i\|\|w_j\|}(\sigma_i\|\Delta A\| + \sigma_j\|\Delta A\| + \|\Delta A\|^2) \\
&\leq \frac{1}{(\sigma_i - \epsilon)(\sigma_j - \epsilon)}(\sigma_i\|\Delta A\| + \sigma_j\|\Delta A\| + \|\Delta A\|^2) \\
&\leq \frac{1}{(\sigma_p - \epsilon)^2}\big[(\sigma_i + \sigma_j)\|\Delta A\| + \|\Delta A\|^2\big] \\
&\leq \frac{1}{(\sigma_p - \epsilon)^2}\big[(\sigma_i + \sigma_j)\|\Delta A\| + \epsilon\|\Delta A\|\big] \\
&= \frac{1}{(\sigma_p - \epsilon)^2}\big[(\sigma_i + \sigma_j + \epsilon)\|\Delta A\|\big] \in O(\|\Delta A\|).
\end{aligned} \tag{4.5}$$

This completes the proof. □

A.3 The solution of an orthogonal Procrustes problem

Claim. The solution of the minimization problem (3.9) is $Q = U'(V')^T$, where $U'\Sigma'(V')^T$ is an SVD of the matrix $\hat{S}\Lambda$.

Proof. Let $Q \in \mathbf{R}^{p\times p}$ be an unitary. Consider

$$\begin{aligned}
\left\|\hat{S} - Q\Lambda\right\|_F^2 &= \langle \hat{S} - Q\Lambda, \hat{S} - Q\Lambda\rangle_F \\
&= \|\hat{S}\|_F^2 - \langle \hat{S}, Q\Lambda\rangle_F - \langle Q\Lambda, \hat{S}\rangle_F + \langle Q\Lambda, Q\Lambda\rangle_F \\
&= \|\hat{S}\|_F^2 - 2\langle \hat{S}, Q\Lambda\rangle_F + \langle Q\Lambda, Q\Lambda\rangle_F \\
&= \|\hat{S}\|_F^2 - 2\langle \hat{S}, Q\Lambda\rangle_F + \|\Lambda\|_F^2 \\
&= \|\hat{S}\|_F^2 + \|\Lambda\|_F^2 - 2\,Tr(\hat{S}^T Q\Lambda) \\
&= \|\hat{S}\|_F^2 + \|\Lambda\|_F^2 - 2\,Tr(\Lambda\hat{S}^T Q) \\
&= \|\hat{S}\|_F^2 + \|\Lambda\|_F^2 - 2\langle \hat{S}\Lambda, Q\rangle_F.
\end{aligned} \tag{4.6}$$

Since $\hat{S}\Lambda = U'\Sigma'(V')^T$, we have

$$\begin{aligned}
\langle \hat{S}\Lambda, Q\rangle_F &= \langle U'\Sigma'(V')^T, Q\rangle_F \\
&= Tr(V'\Sigma'(U')^T Q) \\
&= Tr(\Sigma'(U')^T QV') \\
&= \langle \Sigma', (U')^T QV'\rangle_F.
\end{aligned} \tag{4.7}$$

Note that U' and V' are unitaries, thus the product $R' = (U')^T QV'$ is also an unitary. Combining (3.9), (4.6), (4.7), and the fact that R' is an unitary, it suffices to solve

$$\max_{R' \text{ unitary}} \langle \Sigma', R'\rangle_F = \max_{R' \text{ unitary}} \sum_{i=1}^{p} \sigma_i'(R')_{i,i}. \tag{4.8}$$

Because R' is an unitary, each $(R')_{i,i} \in [-1, 1]$. Hence, to attain the maximum in (4.8), one must take $(R')_{i,i} = 1$ for all i. This implies R' is an identity matrix. Then $I = (U')^T QV'$, thus $Q = U'(V')^T$, as desired. □

A.4 An explicit formula of σ_i' in (3.15)

Claim. $\sigma_i' = \left[\sum_{j=1}^{p} \|\hat{s}_j\|^4 \big((V')_{j,i}\big)^2 \right]^{\frac{1}{2}}, \quad i = 1, \cdots, p.$

Proof. Since $\hat{S}\Lambda = U'\Sigma'(V')^T$, then $(U')^T\hat{S}\Lambda V' = \Sigma'$. This means

$$
\begin{aligned}
(\sigma_i')^2 &= \left\|\Sigma' e_i\right\|_F^2 \\
&= \left\|(U')^T\hat{S}\Lambda V' e_i\right\|_F^2 \\
&= \left\|\hat{S}\Lambda V' e_i\right\|_F^2 \\
&= \sum_{j=1}^{p} \left\|\|\hat{s}_j\|(V')_{j,i}\hat{s}_j\right\|_F^2 \\
&= \sum_{j=1}^{p} \|\hat{s}_j\|_F^2 (V')_{j,i}^2 \left\|\hat{s}_j\right\|_F^2 \\
&= \sum_{j=1}^{p} \|\hat{s}_j\|_F^4 (V')_{j,i}^2.
\end{aligned}
\tag{4.9}
$$

Taking the square root of both sides of equation (4.9) completes the proof. □